



Computer Science Department

Software Engineering (COMP 433) Semester 2 (Spring) 2025/2026

Instructors:

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Introduction:

Software engineering is the discipline concerned with the application of theory, knowledge, and practice for effectively and efficiently building software systems that satisfy the requirements of users and customers. Software engineering is applicable to small, medium, and large-scale systems. It encompasses all phases of the life cycle of a software system.

Software engineering employs engineering methods, processes, techniques, and measurement. It benefits from the use of tools for managing software development; analysing and modelling software artifacts; assessing and controlling quality; and for ensuring a disciplined, controlled approach to software evolution and reuse. Software development, which often involves a team of developers, requires choosing the suitable tools, methods, and approaches that are most applicable for a given development environment.

The elements of software engineering are applicable to the development of software in any computing application domain where professionalism, quality, schedule, and cost are important in producing a software system.

Aims:

To provide an overall understanding of the fundamental concepts, and practical methods for engineering software. It will provide an in-depth understanding of software engineering approaches centred around and grounded into practical context. The course will equip students with foundational knowledge and practical skills to apply software engineering methods and techniques for engineering reasonably large software systems. It will also provide students with analytical means for assessing and evaluating factors that influence the selection and use of appropriate software engineering methods to appreciate their practical applications and their limitations.

Learning Outcomes:

Upon successful completion of this course, students will:

A. Development and Understanding

1. have a good understanding of software engineering foundational concepts
2. have a good knowledge of software engineering techniques and methods
3. be able to demonstrate critical awareness of personal responsibility and professional codes of conduct in the context of software systems development

B. Cognitive/Intellectual Skills

1. have the ability to analyse complex areas of knowledge regarding software engineering
2. know how to critically evaluate software engineering approaches and methods
3. be able to assess and analyse considerations of contextual and circumstantial situations when/where to apply and use software engineering methods and models
4. be able to assess analysis and design considerations to the design of software systems
5. have the ability to recognise and appreciate software engineering technical limitations

C. Key/transferable Skills

1. be able to manage their own learning and information in support of their study
2. know how to undertake self-evaluation of their work
3. be able to work effectively within a team and support the various team functions
4. be able to document and present their own learning and work outputs in suitable written formats.

D. Practical Skills

1. be able to apply software engineering modelling methods (UML) and software system tools
2. be able to practice the stages of software development life cycle through a team project.
3. be able to practice applying software engineering modelling methods and approaches through a team project.
4. be able to practice the presentation of their own learning and work outputs.

Syllabus:

Introduction and background to Software Engineering	1	Chapter 1 (R1)/ (R2) (+ other introductory material)
Software processes, models and techniques (SDLC)	2	Chapter 2 (R2) [+Chapter 3 (R2)]
Software Requirements Engineering/Elicitation	3	Chapter 4 (R1) / (R2)
System Modelling and Analysis Using Unified Modelling Language (UML)	6	Chapter 5 (R2) [Chapter 2 (R1) +Chapter 5 (R1)] (+ UML 2.* from (R3, R4))
System/architectural Design	2	Chapter 6 (R1) + Chapter 7 (R2)
Detailed Design	1	[Chapter 6 (R2) + Chapter 7.1 (R2)]

Teaching and learning method:

- A. Lecture: Theoretical knowledge and practical aspects of the topic will be covered into a series of lectures that provide the foundational principles and concepts.
- B. Tutorial: Understanding of software engineering techniques and methods will be improved through applied case studies, to critically analyse their strengths and weakness, in small group discussions.

- C. Project: a group project as case study of a software application, to apply learned software engineering techniques and methods, to consider a process model, analyse requirements, consider different design goals and factors, represent/model and design software and system aspects, and develop and use engineering techniques to create a complete software model, produced as a UML model and software artefacts.

Assessment methods based on outcomes:

1. Exams to assess: A1, A2, B1, B2, B3, B5
2. Project to assess: A3, B1, B2, B3, B4, C1, C2, C3, C4, D1, D2, D3, D4
3. Exercises to assess: B1, B3, B4, B5, C2, D1

Course Assessment in Normal situation (i.e., Final & Midterm exams are both conducted Face-to-Face):

Formative Assessment:

Mid Term Exam	25%
Group Project, Assignments//Exercises	40%

Summative assessment:

Final Exam	35%
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Course Assessment in other abnormal situations:

Midterm Exam and/or Final Exam Not Face-to-Face	
Mid Term Exam:	25%
Group Project/Exercises:	45%
Final Exam:	30%

Notes:

- The group project is a substantial and essential part of the course and carries significant part of the course assessment. It requires you to form a group of 3-4 of your peers in the first two weeks of the course.
- The project has multiple phases to complete and each requires submission in a written format. Timely project submissions are accepted only.
- Professional codes of conduct are part of the key aims of this course, thus strive for a professional undertaking and presentation of your project work and submissions.
- The course requires using a UML modelling tool, there are several options:
 - Rational Rose UML tool is available, for students with an educational license, to develop UML diagrams. Enterprise Architecture is my favourite UML authoring tool, however, using any other UML authoring tool is acceptable.
 - Two examples of UML drawing tools that offer free educational licenses are included below. However, these are essentially drawing tools and not fully UML compliant tools, i.e., they would draw UML diagrams but they will not enforce or conform to its grammar [make sure to register through the below links and use your official university email]
 - 1- Lucid.app/lucidchart: <https://help.lucid.co/hc/en-us/articles/360049831771-Sign-up-for-a-free-Educational-account>
 - 2-Astah: <https://astah.net/products/free-student-license/>

- The use of AI-based UML modelling tools for AI-generated UML diagrams is NOT Allowed. AI-generated work or diagrams will receive ZERO, and may warrant a transfer to a disciplinary committee.

References:

- R1. Bruegge and Dutoit, ***Object-Oriented Software Engineering Using UML, Patterns, and Java***, 3rd Edition, Prentice Hall, 2013.
- R2. Sommerville I. (2004-2010) ***Software Engineering***, 9th Edition, Addison-Wesley, Harlow, Essex, UK (older editions can also be suitable for this course)

Recommended books and Readings:

- R3. Seidl, Martina, Marion Scholz, Christian Huemer, and Gerti Kappel (2015), ***UML @ Classroom: An Introduction to Object-Oriented Modeling*** Springer. ISBN 978-3-319-12741-5
- R4. OMG (March 2015), **OMG Unified Modeling Language Version 2.5**. Available Jan 2025, <http://www.omg.org/spec/UML/2.5>
- R5. Jeffrey A. Hoffer, Joey F. George, Joseph S. Valacich. (2005) ***Modern System Analysis and Design*** 4th - 7th Edition, (2013), Prentice Hall.
- R6. Roger Pressman, ***Software Engineering: A Practitioner's Approach***, 7-8th edition, McGraw-Hill; 2014.
- R7. L.A. Maciaszek, ***Requirements Analysis and System Design: Developing Information Systems with UML***, 1-3rd Edition, Addison Wesley, 2007.

Dear Students,

Some students were asking about the availability of UML modelling tools, in addition to the ones announced previously, it came to my attention that some online tools (usually more helpful for online team work) offer free educational licenses. Below are two examples, should you wish to use any, if suitable. However, these are essentially drawing tools and not fully UML compliant tools, i.e. they would draw UML diagrams but they will not enforce or conform to its grammar.

1- Lucid.app/lucidchart

<https://help.lucid.co/hc/en-us/articles/360049831771-Sign-up-for-a-free-Educational-account>

2-Astah

<https://astah.net/products/free-student-license/>

However, make sure to register through the above links and to use your official university email.

Equally, we, as teachers, would like to hear feedback from you, should you have had good or bad experience with any of the tools.

Best, Adel